

Gebran Kastoun

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EDUCATION

Cornell University

Ithaca, NY

BS in Electrical and Computer Engineering, Computer Science (GPA: 3.64/4.0)

May 2028

- **Relevant Coursework:** Embedded Systems, Systems Programming, Computer Architecture, Digital System Design with Microcontrollers, Digital Logic & Computer Organization, OOP & Data Structures, Circuits, Microelectronics

EXPERIENCE

SpaceX

May 2026 – Aug. 2026

Avionics Hardware Intern

Hawthorne, CA

- Served as Responsible Engineer for a flight-critical reentry-vehicle avionics box, owning development from requirements through integration and resolving electrical, mechanical, manufacturing, and software interface issues
- Designed two 4-layer PCB interface cards for flight avionics with point-of-load power regulation and differential signal interfaces, integrating replacement flight hardware at 80% lower cost
- Owned testing for an internally developed Xpedition MCP Server, improving schematic design time by 25% and reducing peer review time by 40%; presented findings to 250 employees, garnering 40 new users in 2 weeks

Cornell CUAir – 2nd Place, 2025 C-UASC Competition

Nov. 2024 – Present

Electrical Subteam Lead & Aircraft Pilot

Ithaca, NY

- Developed C++ firmware for closed-loop gimbal stabilization using I²C IMU feedback, PID control, and PWM servo actuation, streaming live angle telemetry over UART through a Raspberry Pi to the ground station
- Designed and brought up an ATmega328P embedded system while debugging hardware/software interfaces, reducing system size and weight by 50% and delivering CUAir's first fully functional integrated gimbal in 6 years
- Own aircraft avionics architecture while mentoring 12 electrical members, reviewing system designs, managing 9 project timelines, coordinating hardware/software integration, and directing & piloting test flights

Cornell University

Aug. 2026 – Present

Teaching Assistant

Ithaca, NY

- **ECE 4760 – Digital Systems Design Using Microcontrollers:** Support 80 undergraduate and MEng students in 8 weekly lab hours, debugging RP2350 systems in C across peripheral interfaces, real-time firmware, concurrency, and hardware-software integration; guide system-level testing and real-time task coordination

PROJECTS

Autonomous Drawing Car | *RP2040, C, Pico SDK, Protothreads, UDP | Circuit Cellar Magazine, Oct. 2026*

- Developed concurrent C firmware across two Pico W microcontrollers using cooperative Protothreads to coordinate USB input, UDP communication, IMU sensing, and motor control; dedicated a second core to VGA rendering
- Used semaphore signaling to defer UDP payload processing from receive callbacks to a dedicated task
- Integrated BNO055 IMU feedback over I²C, PWM motor control, and state-machine navigation to execute wirelessly transmitted waypoints with feedback-based heading control
- Implemented nearest-neighbor route planning with 2-opt refinement to reduce travel distance; deferred optimization until transmission and cached routes, invalidating results when waypoints changed to avoid unnecessary recomputation

Real-Time Optimized Galton Board Simulator | *RP2040, C, PIO, DMA, Multicore, VGA, Fixed-Point*

- Scaled a real-time physics simulator from 200 to 2,000 simultaneous balls at 60 FPS, increasing supported simulation load 10× while maintaining VGA rendering and collision dynamics
- Parallelized physics updates across both RP2040 cores and overclocked the processor to increase simulation throughput
- Replaced floating-point collision calculations with fixed-point arithmetic and alpha-max-beta-min distance approximation; offloaded VGA generation to PIO and DMA, freeing CPU time for physics updates and sustaining 60 FPS under load

TinyRV2 Five-Stage Pipelined Processor | *Verilog, Python, pytest, Linux*

- Implemented stalling and fully bypassed five-stage processors for a 27-instruction TinyRV2 RISC-V subset, extending datapath and control logic to execute compiled C programs with arithmetic, memory, and control-flow operations
- Implemented execute/memory/writeback forwarding, load-use stalls, branch squashing, a variable-latency integer multiplier, and valid/ready instruction and data memory interfaces with backpressure handling
- Developed directed and randomized assembly tests in Python/pytest covering instruction semantics, operand dependencies, control flow, and memory delays; verified both RTL designs against a functional ISA model and used cycle-level traces to benchmark CPI, identify dependency stalls, and evaluate forwarding

SKILLS

Programming Languages & HDL: C/C++, Rust, Python, Verilog/SystemVerilog, Assembly, Java

Systems & Software: Linux, Git, Docker, Real-Time Systems, Multicore Programming, Concurrency, pytest

Embedded Platforms & Interfaces: RP2040/RP2350, ATmega328P, Raspberry Pi, Pico SDK, I²C, UART, PWM, PIO

Hardware & Tools: Altium, Xpedition, Oscilloscopes, Logic Analyzers, DMMs, Hardware/Software Integration

Interests: Competitive Swimming, Distance Running, Skiing, Scuba Diving (PADI), Sailing, FPV Drones, Drums